

Ananth Sriram

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science + Machine Learning

College Park, MD

May 2026

EXPERIENCE

Capital One

Software Engineering Intern

Jun 2025 – Aug 2025

Richmond, VA

- Developed and **productionized** a **zero-touch** pipeline using **AWS Lambda** and **Llama 4 Scout** to automate internal issue triaging; replaced a legacy SVM system and its 3-hour manual CSV workflow with a scheduled **Jira-to-Model retraining loop**, increasing bot utility from a 'file a ticket' prompt to active issue resolution.
- Developed a containerized Selenium module to programmatically execute data uploads via **AWS ECS Fargate** and **Docker**, diagnosing and resolving **10+ critical Docker runtime failures** related to memory constraints and architecture mismatches.
- Orchestrated secure authentication by integrating **AWS Secrets Manager** for service account credential rotation, ensuring SOC2-compliant access to internal admin interfaces.

Columbia University - CRIS Lab

Undergraduate Research Assistant

Sep 2025 – Present

New York City, NY

- Engineered a **stateless reinforcement learning** algorithm using the **Multi-Armed Bandit (RL)** approach with the **UCB1 algorithm** for the **AI-DARWIN genetic framework**, dynamically balancing operator allocation to achieve a 53.3% success rate in ground-truth mechanism recovery.
- Integrated a **Page-Hinkley (PH) reset test** to detect population regime shifts and engineered a **Jaccard-based diversity reward** to **prevent premature convergence**, sustaining elite structural exploration.

University of Maryland - Precision Agriculture Lab

Undergraduate Research Intern

Dec 2023 – Present

College Park, MD

- Developing a multi-factor analytical framework that **integrates and processes a decade of geospatial and tabular data** (Sentinel-2, NASS) using **Google Earth Engine** and Python, showcasing proficiency in **large-scale data ingestion**.
- Co-authored "An Overview of Drones in Agriculture" (**Second Author**), survey paper for Machine Learning applications for drone-based precision agriculture.
- Currently under revision for the Journal of Applied Earth Observation and Geoinformation (**First Author**); novel satellite-imagery based framework for stress analysis in Maryland.

ZoomInfo

Machine Learning Engineer Intern

Jun 2024 – Aug 2024

Bethesda, MD

- Built a **scalable AutoML SDK** from scratch using Python (*scikit-learn*, *PyTorch*) that architected **batch data processing workflows** to automate model selection and evaluation across **high-volume datasets**.

PROJECTS

1st Place, \$10K Winner | Ironsite.ai (SPC W24) Spatial Intelligence Hackathon, UMD

Feb 2026

- Built a 3-stage ML pipeline (**YOLO11** → **SAM3** → **method-prompted Qwen3-VL-8B with 3-pass adversarial verification**) to automatically detect PPE non-compliance and proximity hazards from egocentric construction footage, mapping violations to OSHA standards with severity scoring; work **published** as an **arXiv preprint (First Author, endorsed by Dr. Ramani Duraiswami)**.
- Designed a persona-scaffolded adversarial chain-of-thought **VLM** protocol with structurally enforced pass independence, achieving a **12% precision improvement** over single-pass prompting on the development corpus.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Go, SQL, OCaml, Ruby

Cloud/Infrastructure: AWS (ECS Fargate, Lambda, ALB, SNS), Docker, Linux, HPC, Git

Backend/Data: Flask, Batch/Stream Processing, Large-Scale Data Analysis, ETL Pipelines

AI/ML Tools: PyTorch, TensorFlow, scikit-learn, AutoML, MLOps, Pandas, NumPy, pytest